

The Volatility Targeted Mean Variance Multilayer Perceptron

Hema Srikar Ankem

Abstract

This document describes a volatility targeted mean variance portfolio for monthly construction, built on a neural forecast of returns. The model separates the problem into a forecast of the conditional mean, a shrinkage estimate of the conditional covariance, and an explicit mean variance optimisation subject to a volatility target and position limits. We develop each component in full and then the optimiser, and relate the design to the finding that on this panel timing the aggregate level of risk helps whereas modelling risk at the level of individual securities does not.

1 Overview and motivation

Selection and risk can be combined in two ways. One approach folds them into a single objective by training a network to output portfolio weights directly. The present model instead follows the classical decomposition of Markowitz, in which an estimate of the expected return vector and an estimate of the return covariance are combined by an explicit optimisation. The purpose is to make the risk model explicit and to impose a volatility target and position limits as hard constraints, so that the leverage of the book is controlled by construction rather than by a post hoc overlay.

2 The conditional mean forecast

The inputs to the network are the centred within month percentile ranks of the retained characteristics, formed under a coverage filter that keeps a characteristic if it is present in at least ninety percent of the pre test observations, with securities missing more than one third of the characteristics dropped. A multilayer perceptron with a pooled contextual term estimates the demeaned next month excess return of each security,

$$\mu_i = \mathbb{E}[r_i - \bar{r} \mid x_i], \quad (1)$$

where $\bar{r}$ is the cross sectional mean and the target is demeaned within the month, so that the forecast is a relative expected return. The pooled contextual term couples the forecast across the cross section, so that a security's forecast reflects the composition of its month. The network is fitted under squared error on a panel of training months, three seeds are fitted and their forecasts are averaged, and the network is refitted once per year on the months whose forward returns precede that year.

3 The shrinkage covariance

Let S denote the sample covariance of the trailing year of daily returns for the securities that have sufficient history. Because the number of securities is large relative to the number of daily

observations, the sample covariance is poorly conditioned, and it is shrunk towards a constant correlation target F in the manner of Ledoit and Wolf. The target retains the sample variances and replaces every pairwise correlation by the cross sectional average correlation, and the shrinkage intensity is

$$\delta = \max\left(0, \min\left(1, \frac{\pi - \rho}{\gamma T}\right)\right), \quad \Sigma = \delta F + (1 - \delta)S, \quad (2)$$

where π is the summed asymptotic variance of the entries of S , ρ is their summed asymptotic covariance with the entries of F , γ is the squared distance between F and S , and T is the number of observations. The daily estimate is scaled to a monthly horizon by the number of trading days in a month.

4 The mean variance problem

Given the forecast μ and the covariance Σ , the traded weights solve

$$\max_w \mu^\top w \quad \text{subject to} \quad w^\top \Sigma w \leq v^2, \quad |w_i| \leq c, \quad \|w\|_1 \leq G, \quad (3)$$

where v is the monthly volatility target corresponding to ten percent per annum, c is a per security cap set from a minimum breadth requirement, and G is a gross exposure limit. The volatility target is a hard risk constraint, the cap enforces diversification by bounding the weight of any one security, and the gross limit bounds the total leverage. The weights are not renormalised to constant gross exposure, so the optimiser is free to scale the book down when the risk constraint binds.

5 The optimiser

The structure of the problem guides the method by which it is solved. For a fixed risk multiplier λ the inner problem is the box constrained quadratic program

$$\max_{|w_i| \leq c} \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w, \quad (4)$$

which is solved by accelerated projected gradient ascent with a projection onto the box $[-c, c]$ at each step. The momentum is restarted whenever it opposes progress, since plain acceleration oscillates on this problem and leaves the iterate short of the precision that the outer constraint requires. The multiplier λ is then chosen by bisection so that the attained volatility meets the target, and the smallest feasible multiplier is preferred, which corresponds to taking as much risk as the target allows. The gross exposure limit is imposed afterwards by a uniform scaling of the solution.

6 Temporal integrity and evaluation

Several safeguards guard against lookahead. The network is refitted yearly on months up to the last month end preceding that year, the covariance uses only the preceding year of daily returns, and the coverage filter that selects the characteristics is recomputed yearly from months at or before the cutoff. On the evaluation window the model attains a Sharpe ratio of 2.753 in the volatility scaled form and 2.912 in the unscaled form.

7 Concluding remarks

The model succeeds in the sense that it outperforms a portfolio that maps characteristics directly to weights without an explicit risk model, and the improvement is consistent with the finding that timing the aggregate level of risk is valuable on this panel. At the same time, the evidence indicates that pushing the covariance information down to the individual security level, through daily estimated beta, idiosyncratic shrinkage, or full covariance mean variance optimisation, tends to reduce performance, because with a cross section substantially wider than the estimation window the per security estimation error does not average out. We can therefore conclude that the value of the risk model here lies primarily in its control of the aggregate volatility of the book rather than in the fidelity of its pairwise covariances, and that the volatility target is the component doing most of the work.